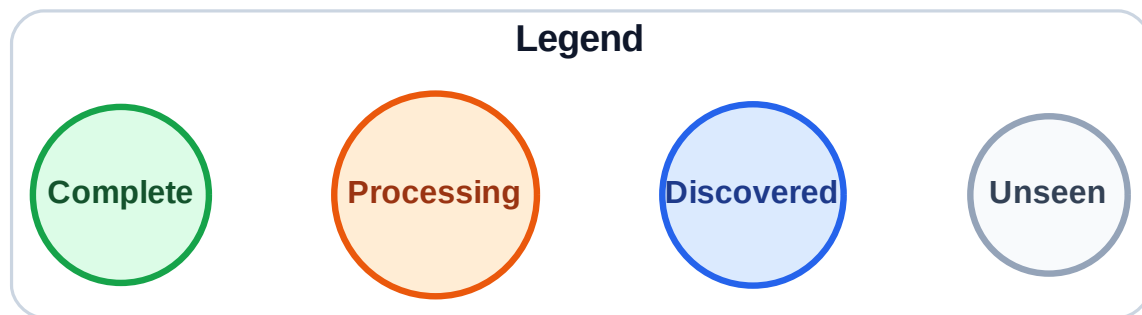
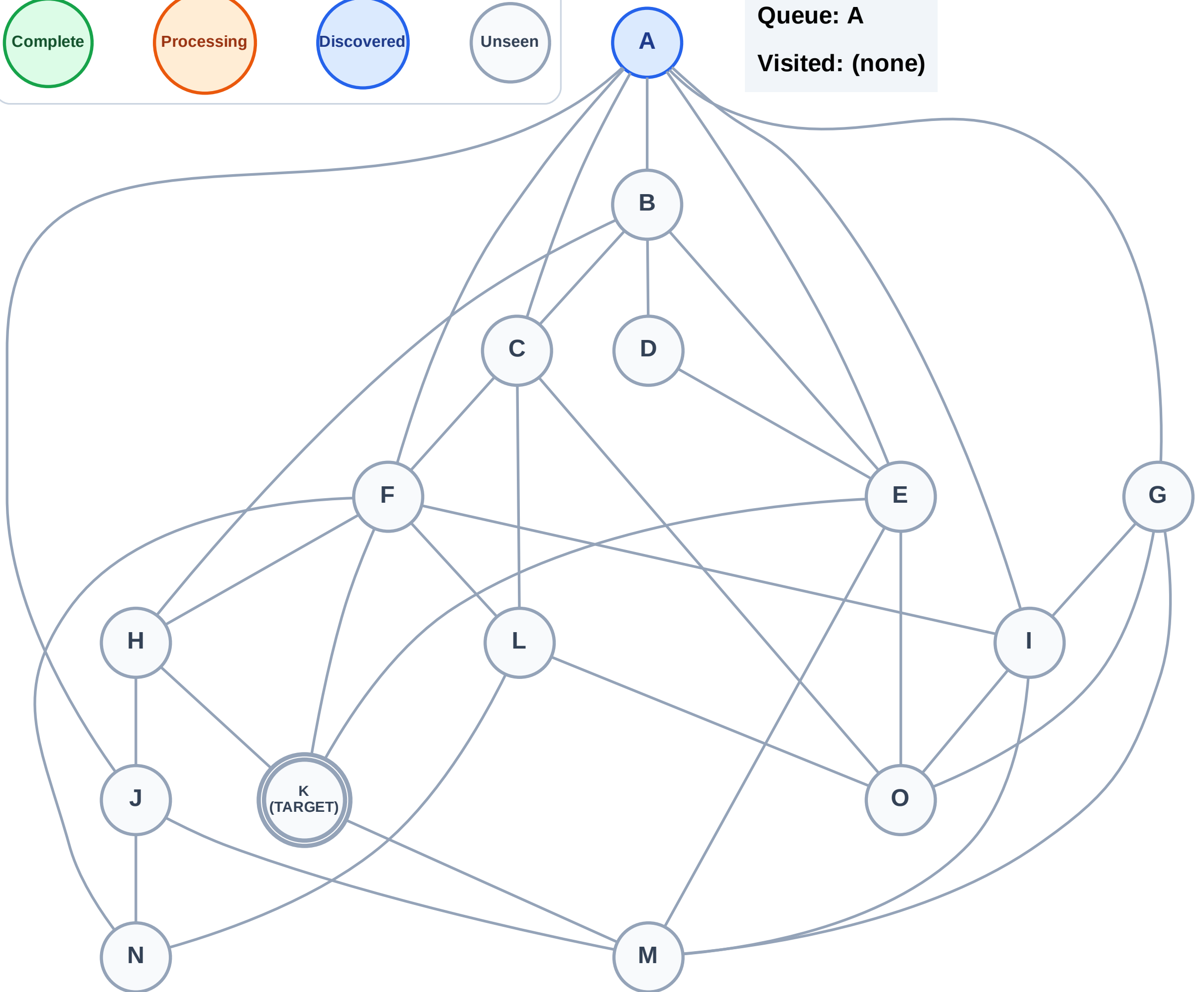


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

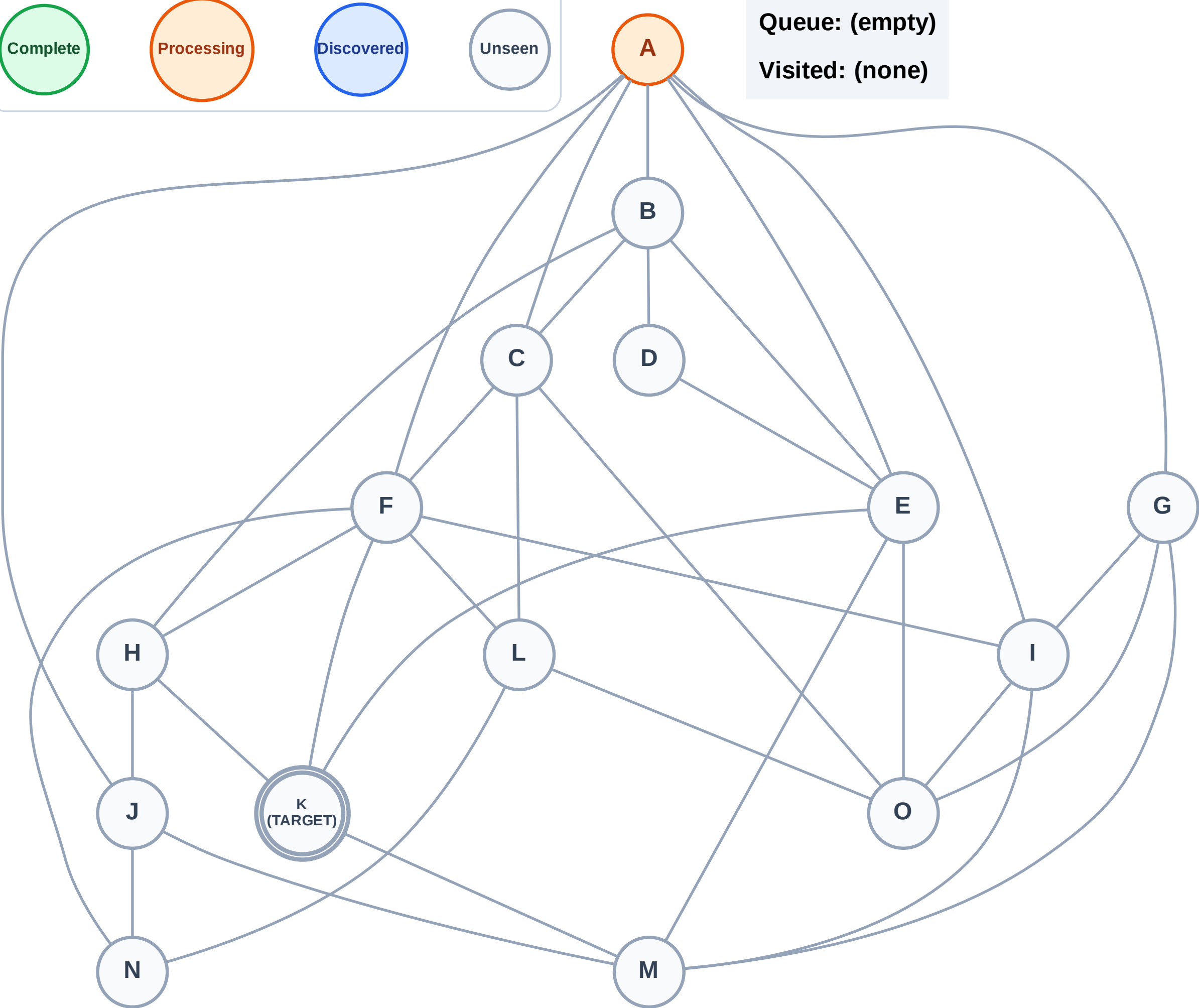
Complete

Processing

Discovered

Unseen

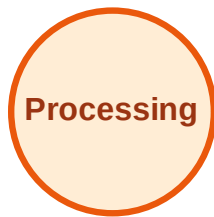
Queue: (empty)
Visited: (none)



BFS Traversal

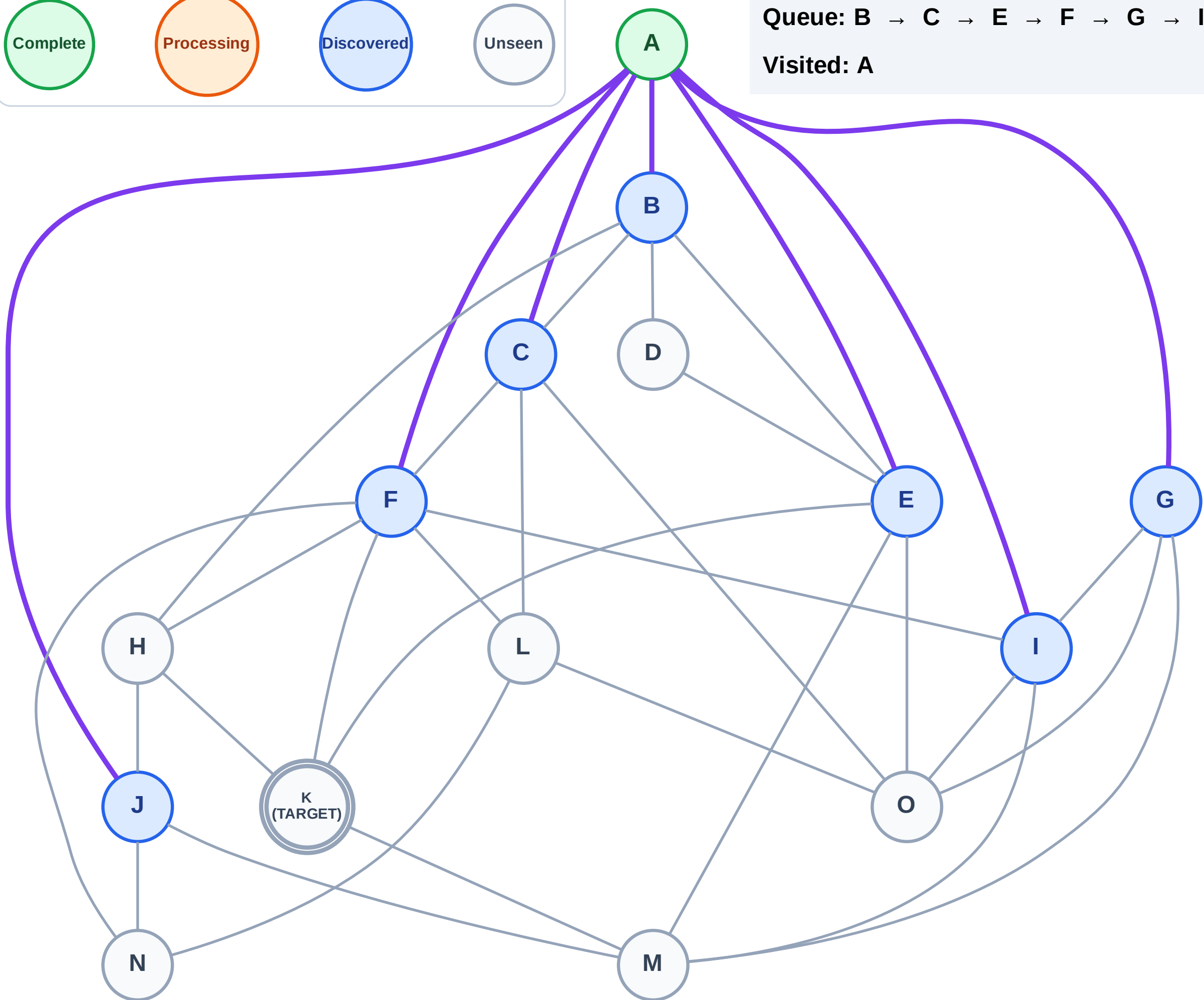
Step 3: Discover B, C, E, F, G, I, J from A

Legend



Queue: B → C → E → F → G → I → J

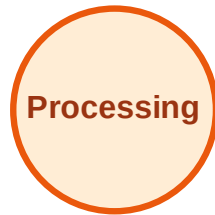
Visited: A



BFS Traversal

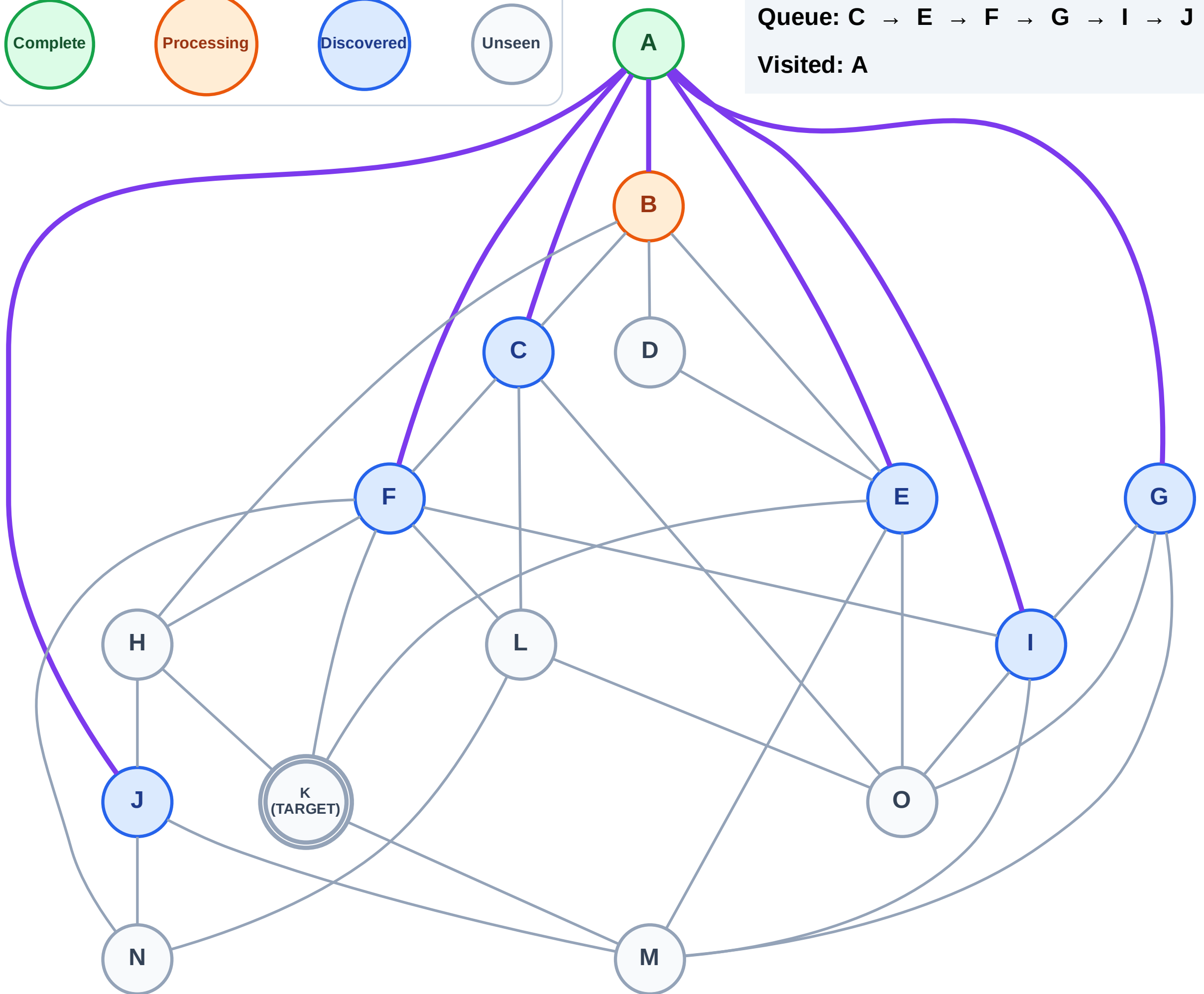
Step 4: Process node B

Legend



Queue: C → E → F → G → I → J

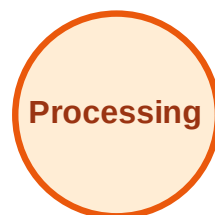
Visited: A



BFS Traversal

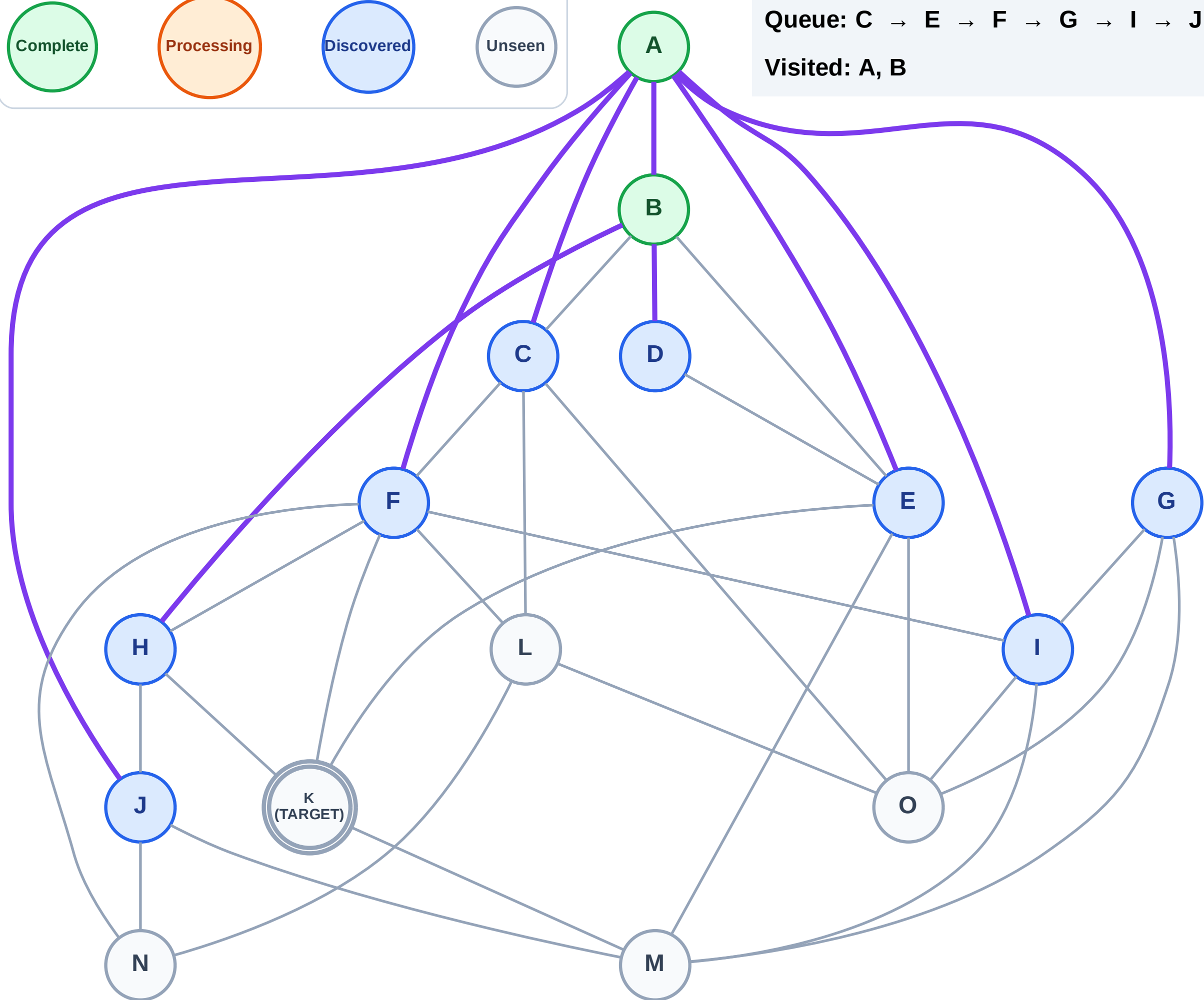
Step 5: Discover D, H from B

Legend

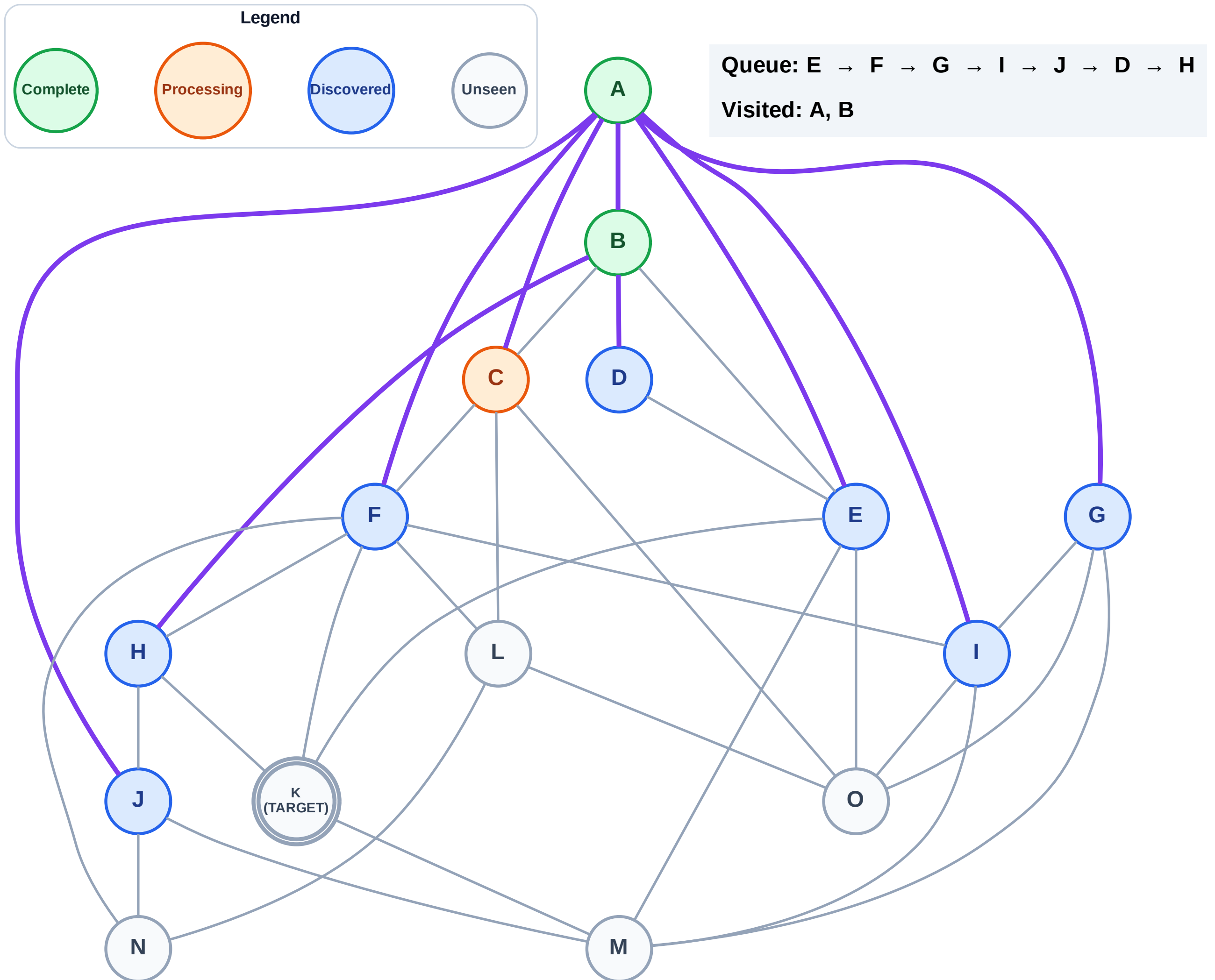


Queue: C → E → F → G → I → J → D → H

Visited: A, B



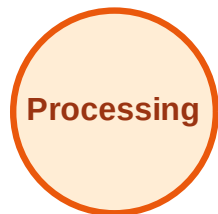
Step 6: Process node C



BFS Traversal

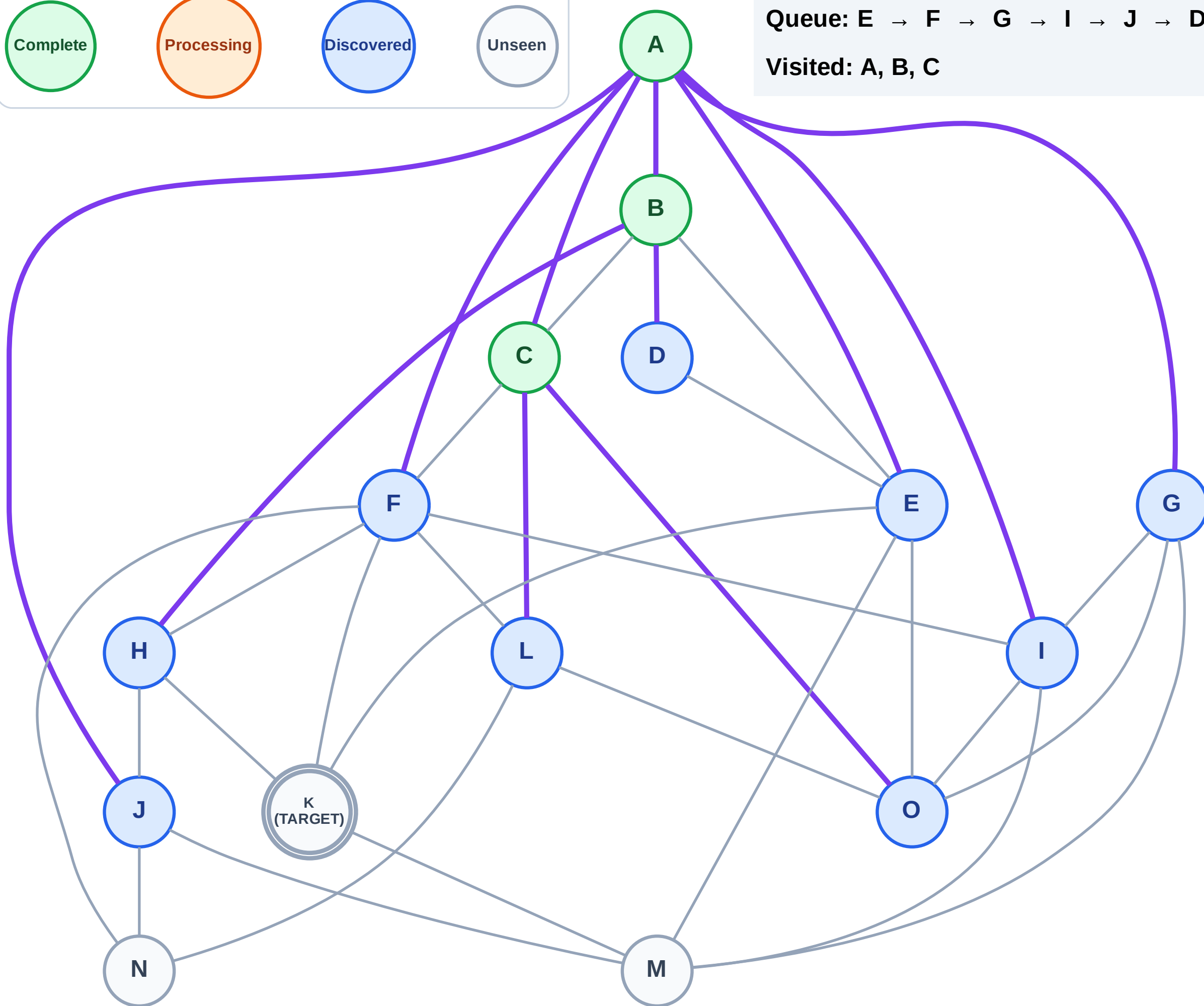
Step 7: Discover L, O from C

Legend



Queue: E → F → G → I → J → D → H → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

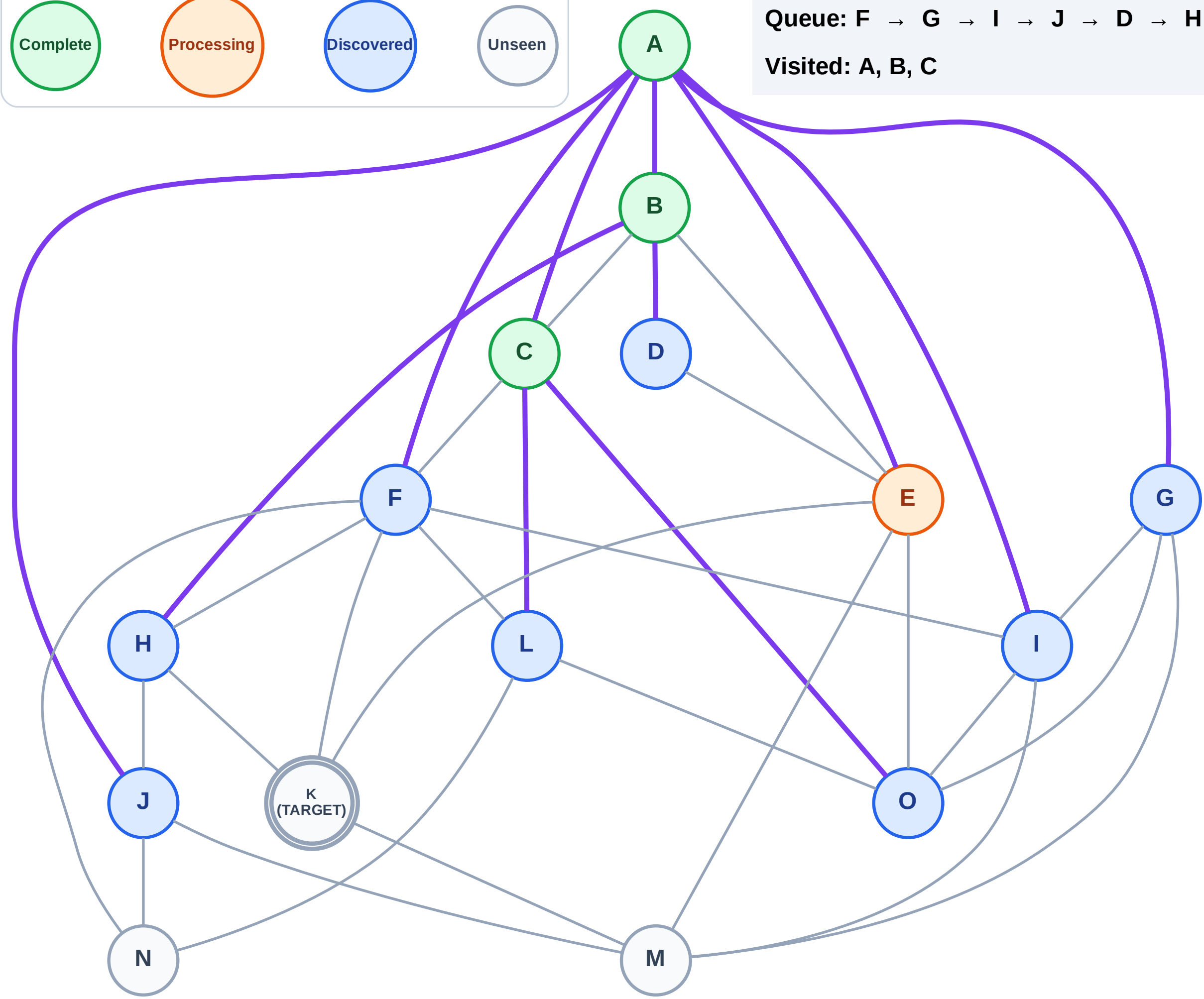
Processing

Discovered

Unseen

Queue: F → G → I → J → D → H → L → O

Visited: A, B, C



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend

Complete

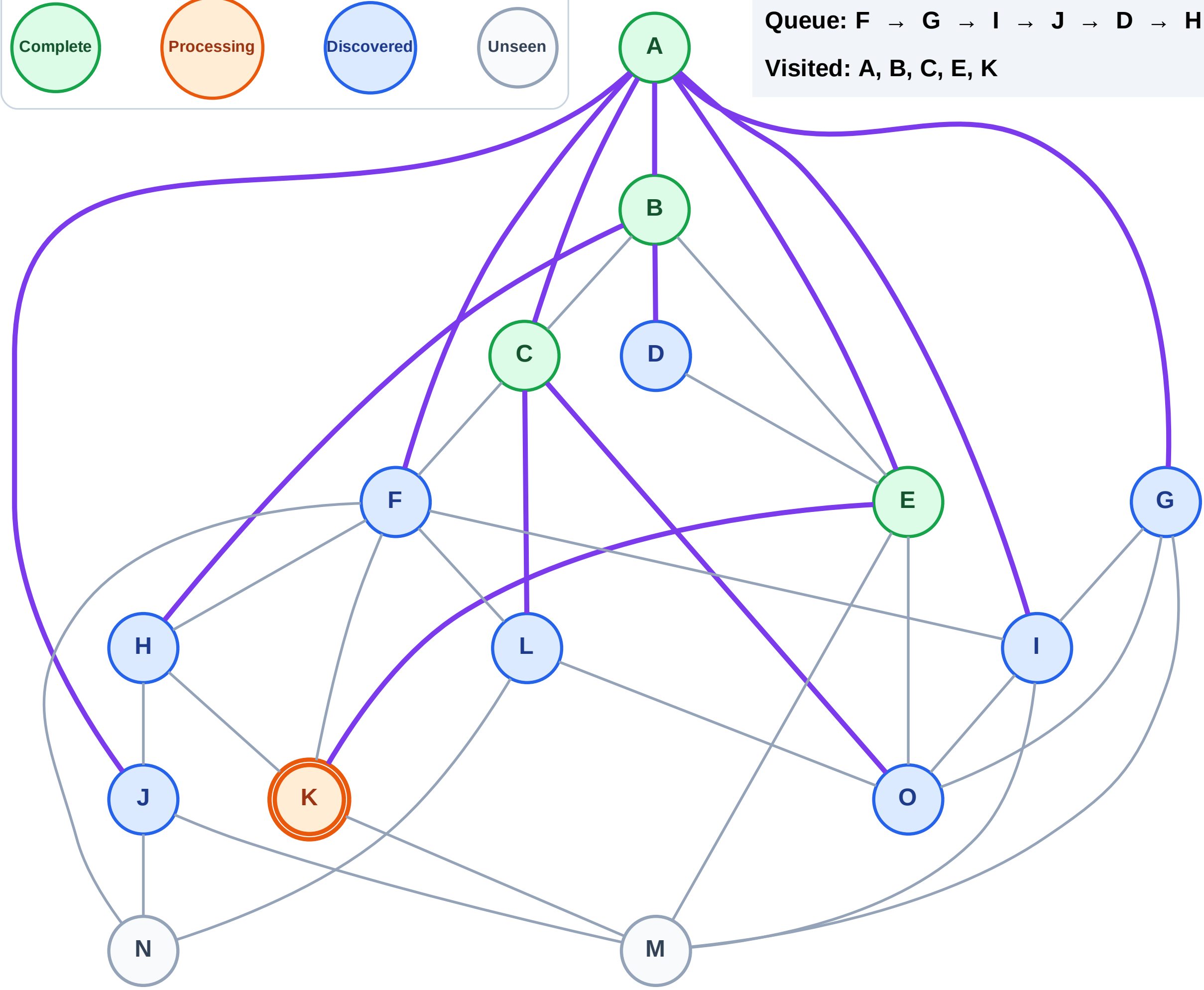
Processing

Discovered

Unseen

Queue: F → G → I → J → D → H → L → O

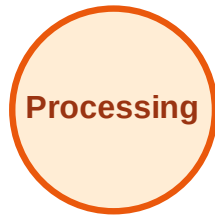
Visited: A, B, C, E, K



DFS Traversal

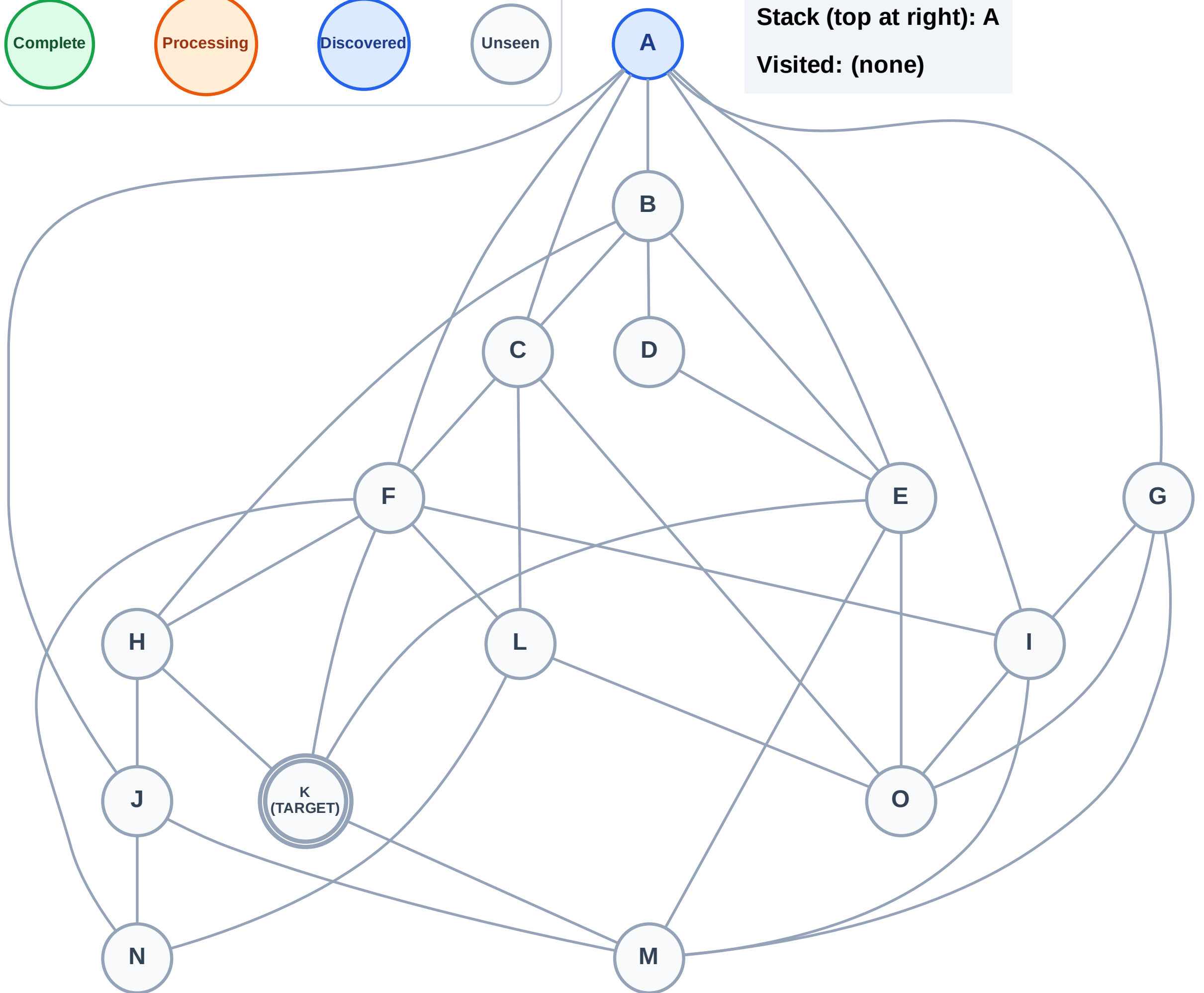
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

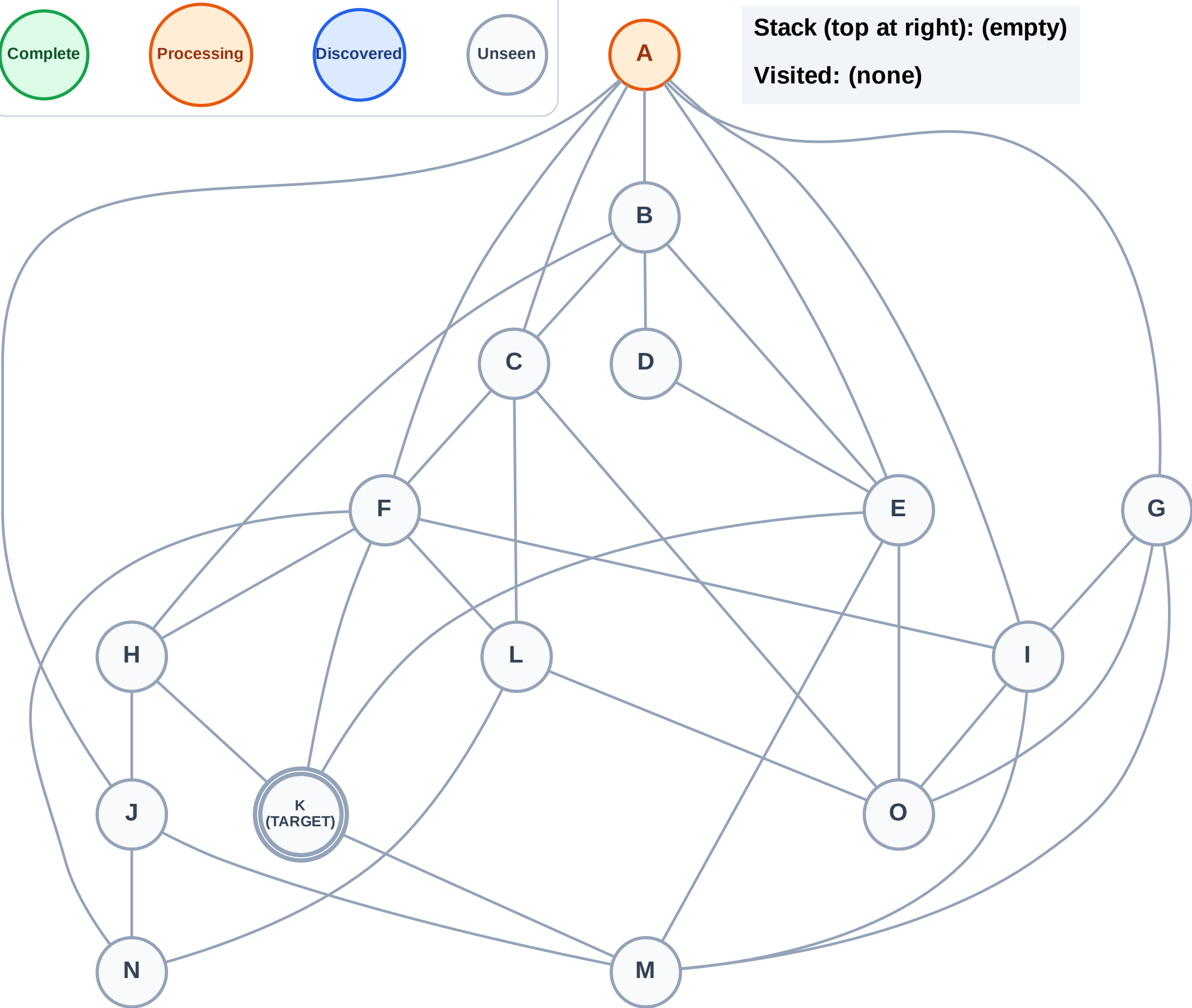
Processing

Discovered

Unseen

Stack (top at right): (empty)

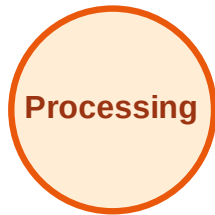
Visited: (none)



DFS Traversal

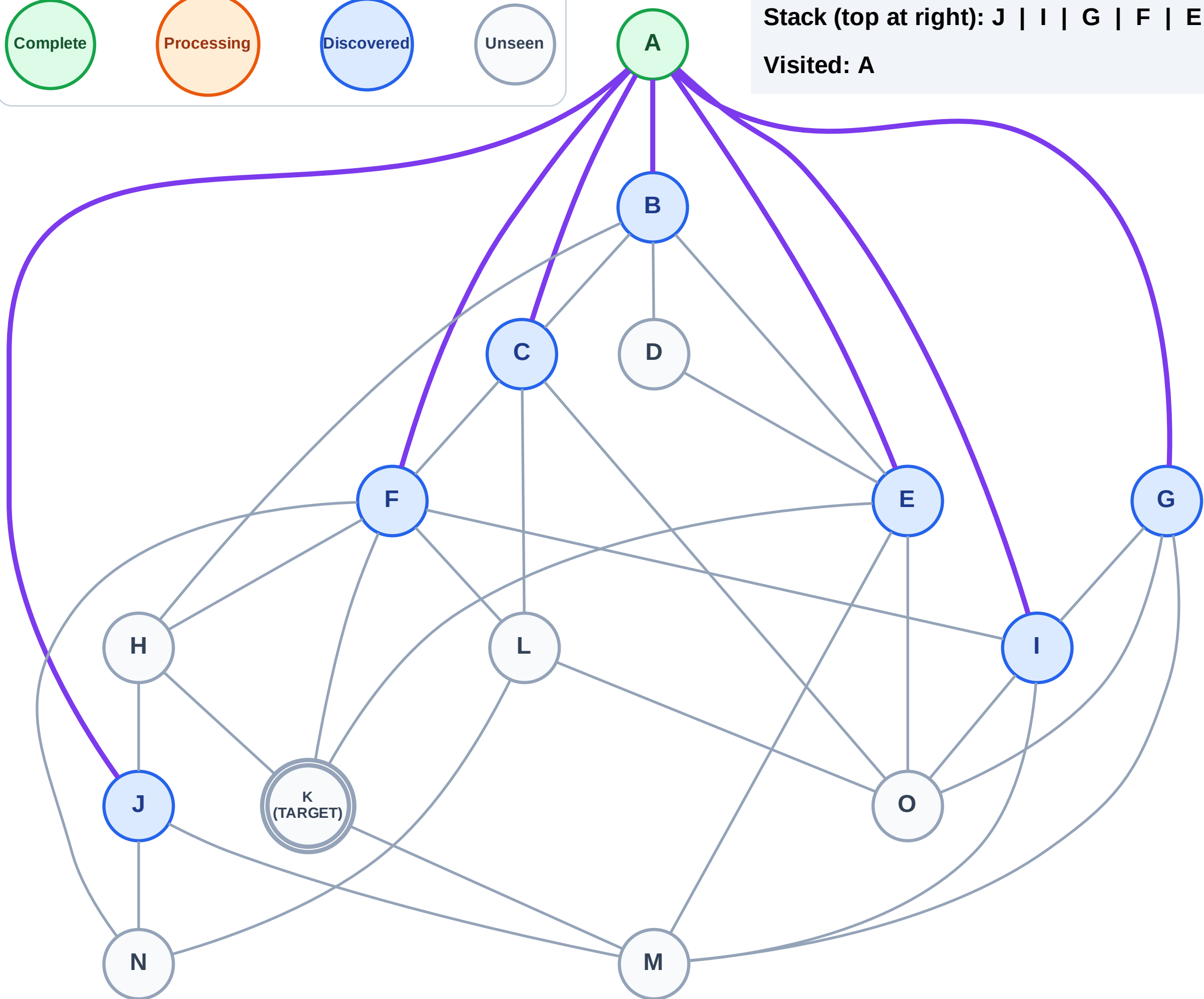
Step 3: Discover J, I, G, F, E, C, B from A

Legend



Stack (top at right): J | I | G | F | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

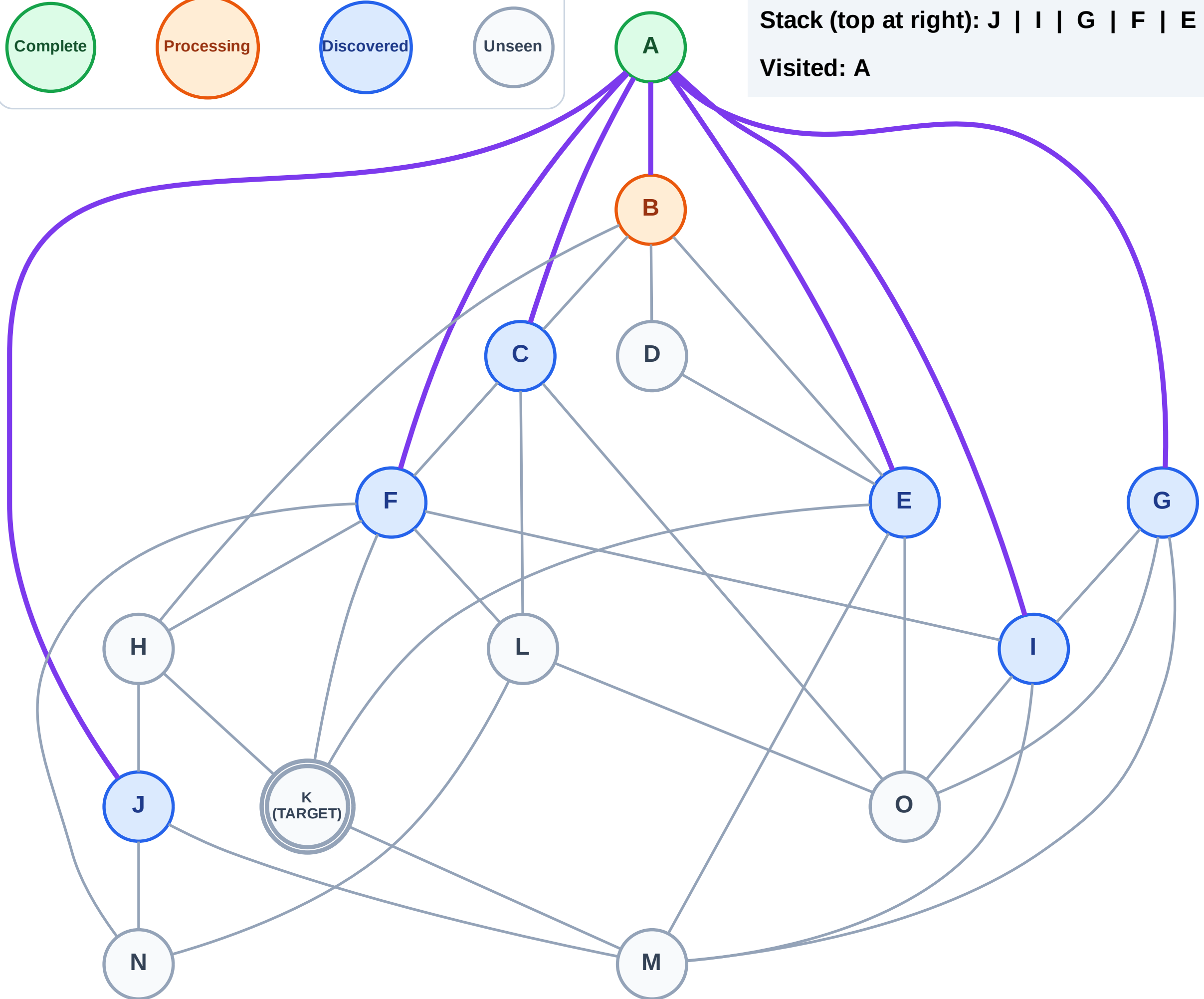
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | E | C

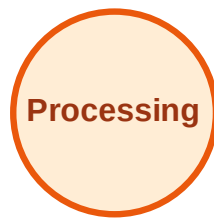
Visited: A



DFS Traversal

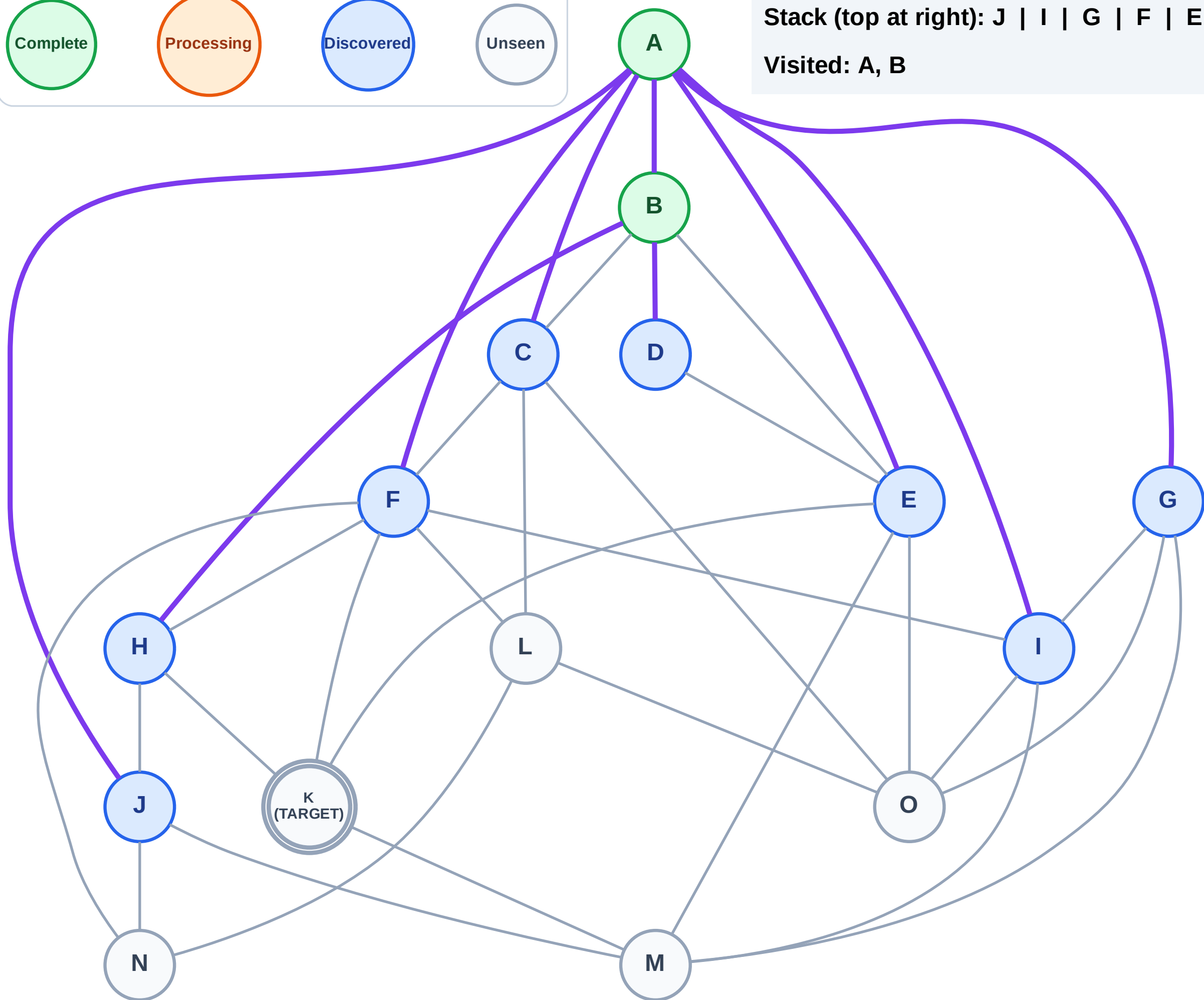
Step 5: Discover H, D from B

Legend



Stack (top at right): J | I | G | F | E | C | H | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

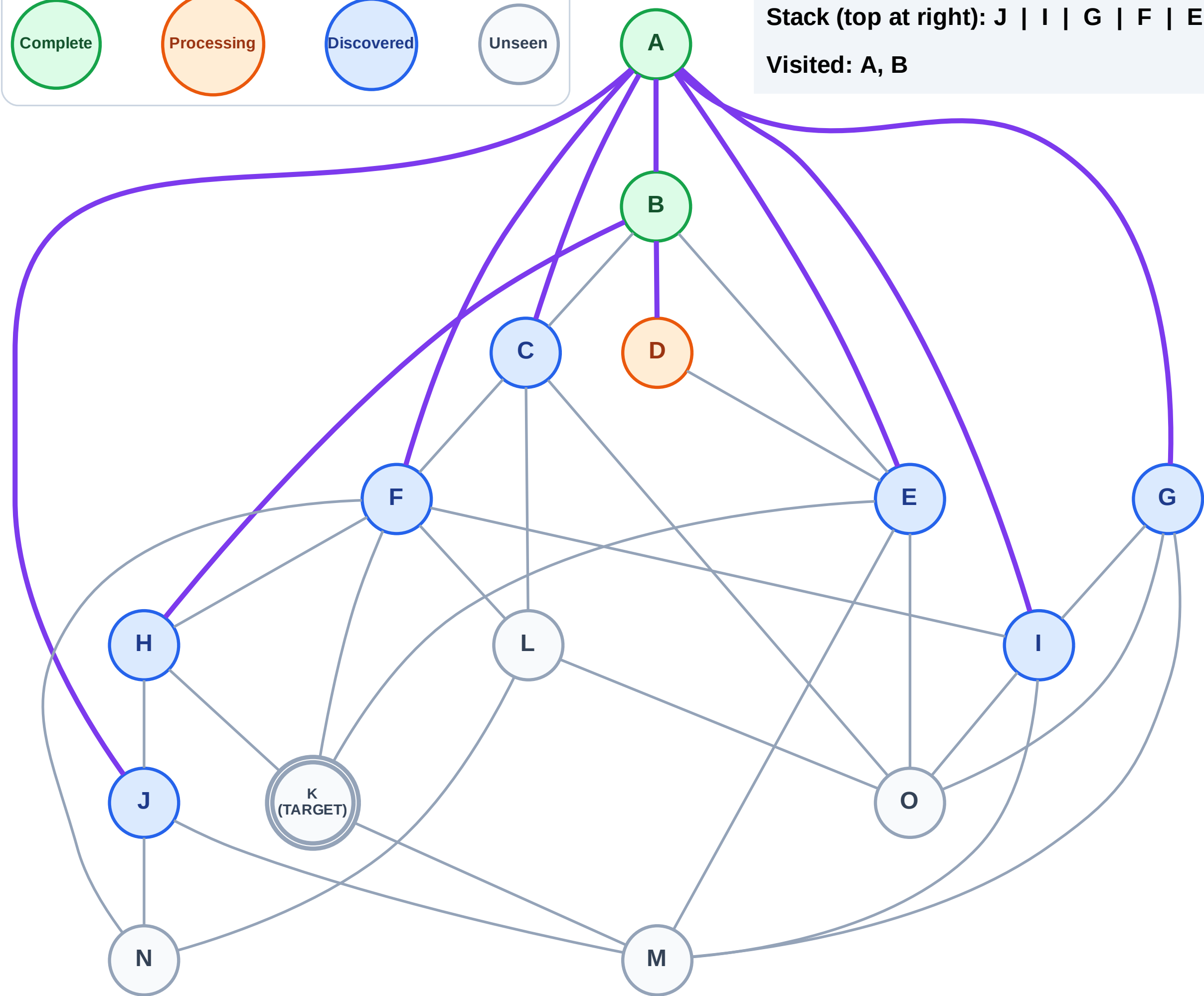
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | E | C | H

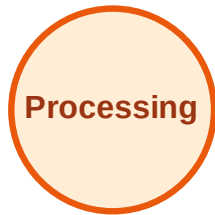
Visited: A, B



DFS Traversal

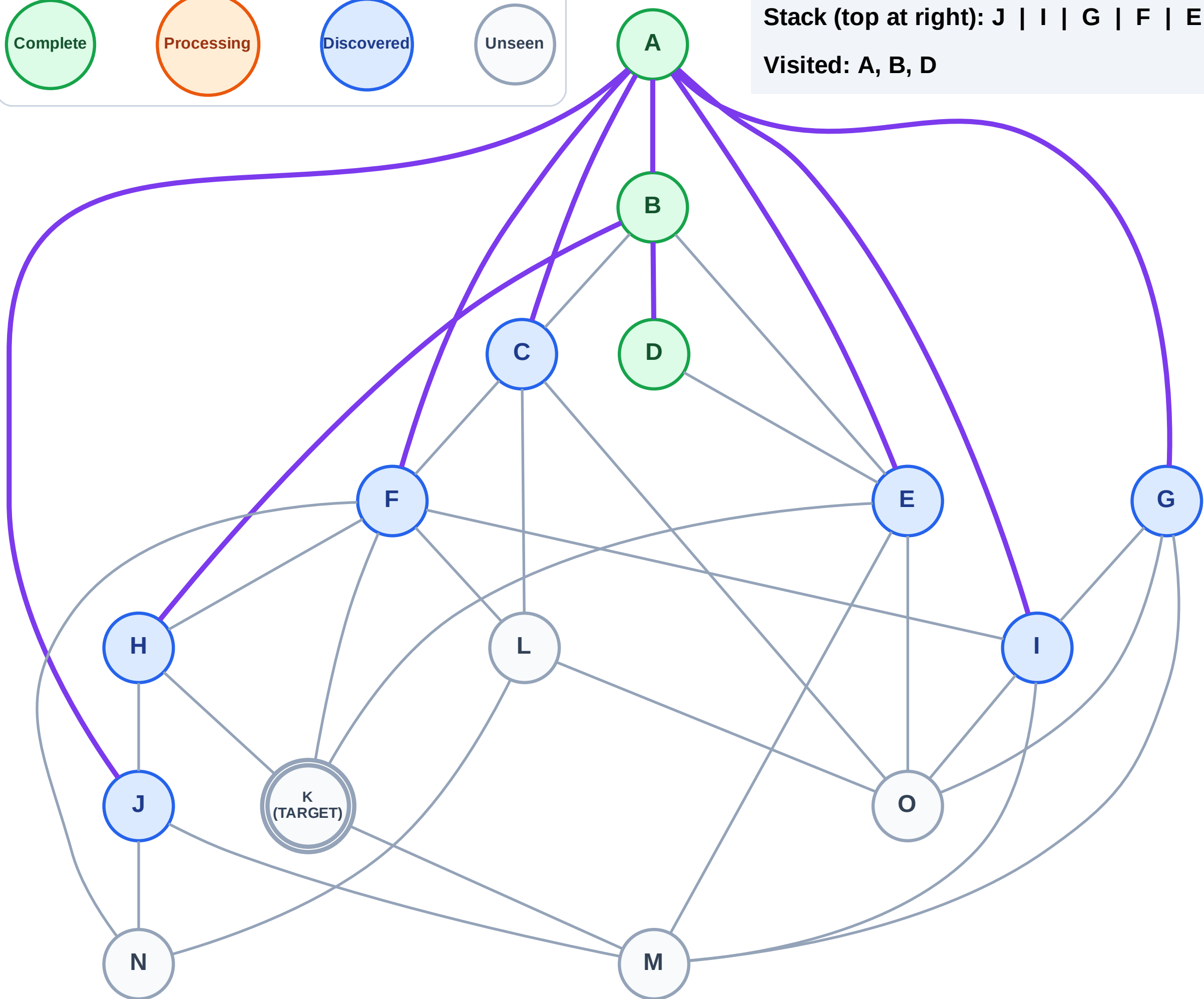
Step 7: No new neighbors from D

Legend



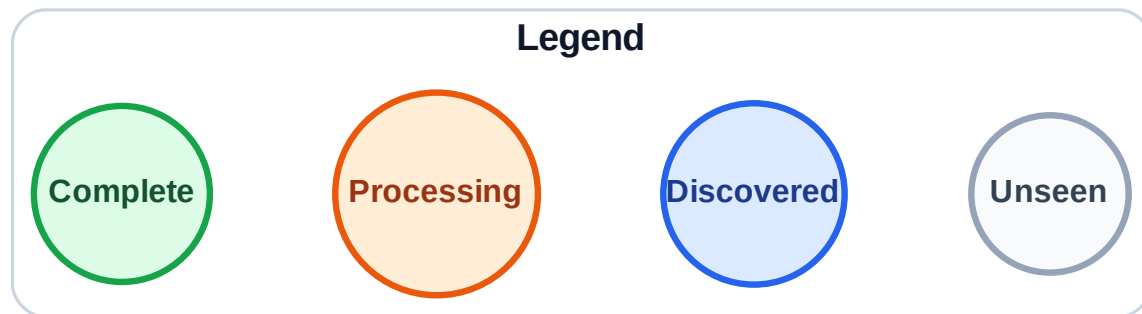
Stack (top at right): J | I | G | F | E | C | H

Visited: A, B, D

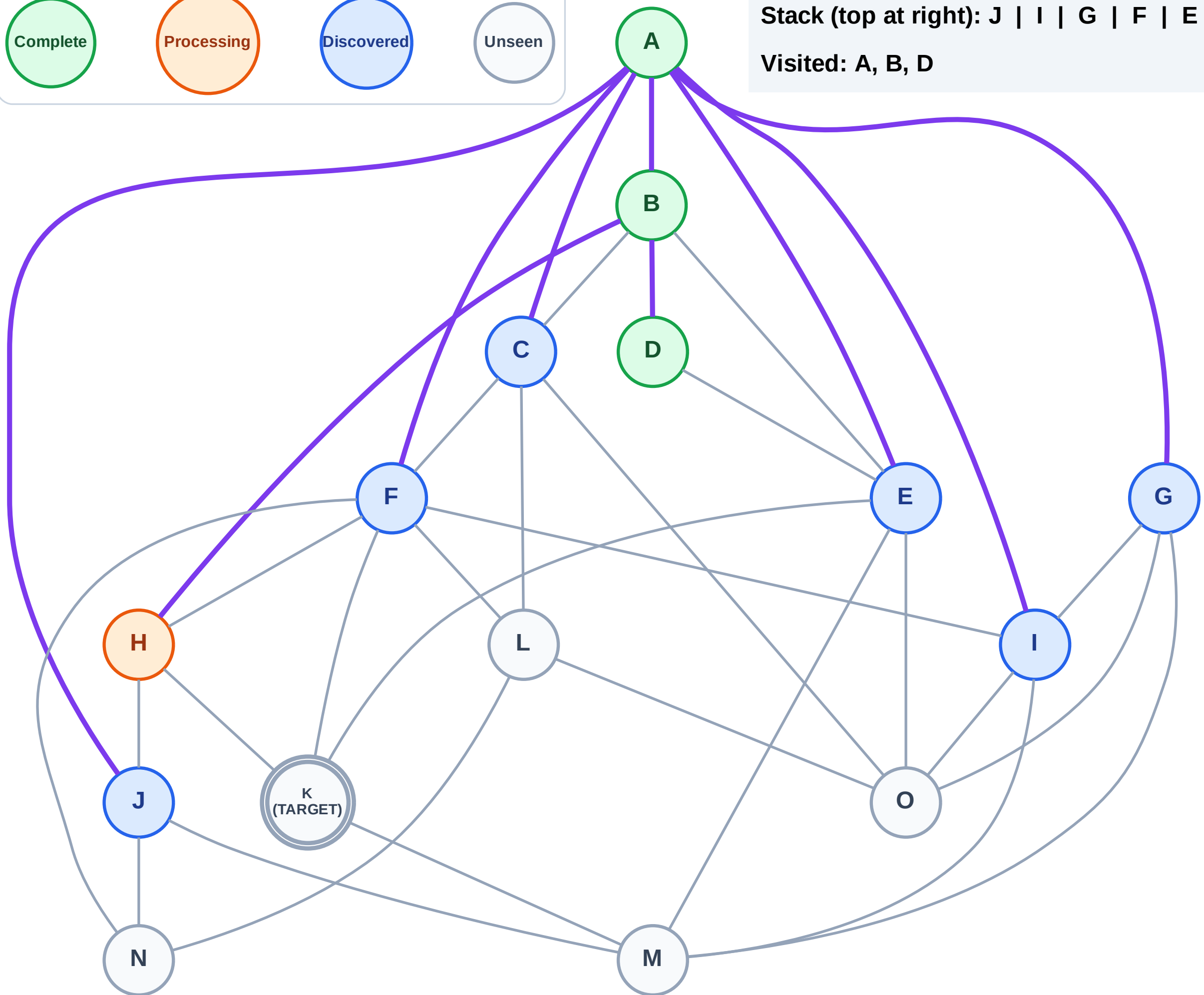


DFS Traversal

Step 8: Process node H



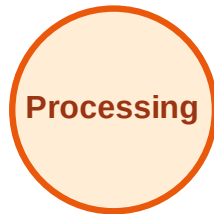
Stack (top at right): J | I | G | F | E | C
Visited: A, B, D



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → H → K

Legend



Stack (top at right): J | I | G | F | E | C

Visited: A, B, D, H, K

